

CS 514: Data Structures and Algorithms-II

Course Code	CS 514
Course Name	Data Structures and Algorithms-II
L-T-P-C	3-0-2-4
Prerequisites	CS-202 (DSA) or equivalent for UG students, None for PG students
Intended for	BTech (Semester 5 or 6) who have not taken CS403, MTech (CSE), MS, PhD
Distribution	Discipline Core for MTech CSE, Elective for UG CSE/DS, MS/PhD
Approval	recommended by the 40th BoA (May 2021)

Course Contents

- **Review of Data Structures from CS 202:** Asymptotic Notations, Solving recurrences, Insertion Sort, Merge Sort, QuickSort. [3 Lectures]
- **Advanced topics in Sorting and Searching:** Randomized Data Structures Treaps, Hashing, AVL Trees, Bucket Sort. [6 Lectures]
- **Dynamic Programming.** [3 Lectures]
- **Amortized analysis:** aggregate analysis, accounting, potential method. [3 Lectures]
- **Graph Algorithms:** Single-Source Shortest Paths Bellman Ford, All-Pairs Shortest Paths using Floyd Warshall, Maximum Flow (Ford Fulkerson). [6 Lectures]
- **Advanced Data Structures:** Quake heaps, van Emde Boas Trees, Union Find Data structures. [6 Lectures]
- **Computational complexity:** Problem classes: P, NP, NP-complete, NP-hard. Reduction. Cooks theorem. Examples of NP-complete problems. [6 Lectures]
- **Approximation Algorithms:** Greedy and Local Search algorithms, DP Algorithms. [3 Lectures]
- **Parameterized Complexity:** Introduction to FPT, Bounded Search Trees, Kernels
- **Additional Topics:** Streaming Algorithms:- Misra Gries, Countmin sketch, LSH, lossy count algorithm. [3 Lectures]
- **Comp. Geometry:** Convex Hull, Line segment intersection, Voronoi Diagrams. [3 Lectures]
- **Coding Lab** which covers topics discussed in this course [28 Hours]

Text Books

1. T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, **Introduction to Algorithms**, 3rd Edition, MIT Press, 2009.
2. J. Kleinberg and E. Tardos, **Algorithm Design**, Pearson, 2006.

References

1. S. Dasgupta, C. H. Papadimitriou, U. V. Vazirani, **Algorithms**, McGraw Hill, 2006.
2. S. S. Skiena, **The Algorithm Design Manual**, 2nd Edition, Springer, 2008.

This course is proposed as a core course for MTech (CSE) and BTech students shall also be allowed to credit the same. The overwhelming overlap of contents (with CS403) is deliberate because we intend to discontinue offering CS403 starting from the winter session of 2021. Some parts of CS403 have either been covered in CS202 or are not relevant to the currently trending research areas. We compensate for these topics by adding new topics which we feel are best suited for the students and this upgrade in coursework also makes this course suitable for M.S., M.Tech and Ph.D. candidates.